

Final Project Proposal

Optimal Siting of Electric Vehicle Charging Stations under a Limited Budget

Problem Description

A city planner has a limited budget and a list of candidate sites where public EV charging stations could be built. Every site costs a different amount to develop, and each station can only serve a limited number of vehicles per day. Drivers are spread unevenly across the city, so a station placed in the wrong area leaves some neighborhoods with a long drive to the nearest charger. The planner has to decide which subset of candidate sites to build, and which station each area of demand should be served by, so that the city gets the best coverage it can afford. This is a capacitated facility location problem. The interesting part is the trade off it contains: building few stations is cheap but forces some drivers to travel far, while building many stations gives good coverage but breaks the budget. The model makes that trade off explicit instead of leaving it to intuition.

Practical Application

Cities and utilities in the United States are actively deciding where to place public charging infrastructure as EV adoption grows, and the same model structure is used for warehouse siting, clinic placement, and cell tower planning. The decision maker here is a municipal planning office or a charging network operator. The output is a concrete build list plus a map, which is exactly the form a planning recommendation takes in practice.

Optimization Method, Tool, and Solver

Mixed integer linear programming (MILP). The binary build decisions make the problem discrete while the assignment decisions are continuous, so the model is solved by branch and bound. The instance is planned to stay small enough, roughly 50 demand points and 20 candidate sites, that an exact solver reaches proven optimality quickly, so no heuristic is required. If the instance is scaled up later and exact solution becomes slow, a greedy construction heuristic will be added as a comparison and that choice will be justified explicitly. The model will be built in Python with PuLP and solved with CBC, matching the workflow used in the course labs. Supporting libraries are pandas and numpy for data handling, osmnx and geopandas for pulling candidate site coordinates, and matplotlib for the map visualizations.

Objective

Minimize the total of construction cost and weighted travel cost:

$$\text{minimize } \sum_j f_j y_j + \lambda \sum_i \sum_j w_i d_{ij} x_{ij}$$

where f_j is the fixed cost of building at site j , w_i is the demand weight (population) at area i , d_{ij} is the distance from area i to site j , and λ converts distance into cost units. A second version of the model will also be run, fixing the budget and maximizing covered demand instead, so the two modeling choices can be compared.

Decision Variables

- $y_j \in \{0, 1\}$: equals 1 if a charging station is built at candidate site j , and 0 otherwise
- $x_{ij} \in [0, 1]$: the fraction of demand at area i that is served by station j

Main Constraints

- Demand satisfaction: for every area i , the fractions assigned across all sites sum to 1
- Linking: $x_{ij} \leq y_j$, so demand can only be assigned to a site that is actually built
- Capacity: for every site j , the total demand assigned to it does not exceed its capacity
- Budget: the total construction cost does not exceed the available budget
- Service radius: x_{ij} is fixed at 0 whenever d_{ij} exceeds a maximum acceptable driving distance
- Non negativity and binary restrictions on the variables

Data Source

Real geography with assumed cost and demand parameters, which will be stated clearly in the report. Candidate site coordinates come from OpenStreetMap through osmnx, using existing parking lots and fuel stations in a single city as realistic locations. Demand points come from overlaying a grid on the city boundary and taking each cell center. Demand weights, construction costs, and station capacities are assigned from documented assumptions, for example a higher construction cost near the city center. Distances are computed directly from the coordinates.

Expected Outcome

A solved model that returns which sites to build, how demand is assigned to them, the optimal objective value, and the solver status. Results will be shown as a map of the chosen stations with their service areas, next to a map of all candidate sites, so the selection is easy to read. The sensitivity analysis will sweep the budget across a range and plot the resulting coverage curve, showing where extra spending stops buying much extra coverage, and will also vary the service radius and the station capacity. For decision support, the model will be wrapped in a repeatable function that takes a budget and returns the recommended build list and map, so the planner can test scenarios without touching the model code.